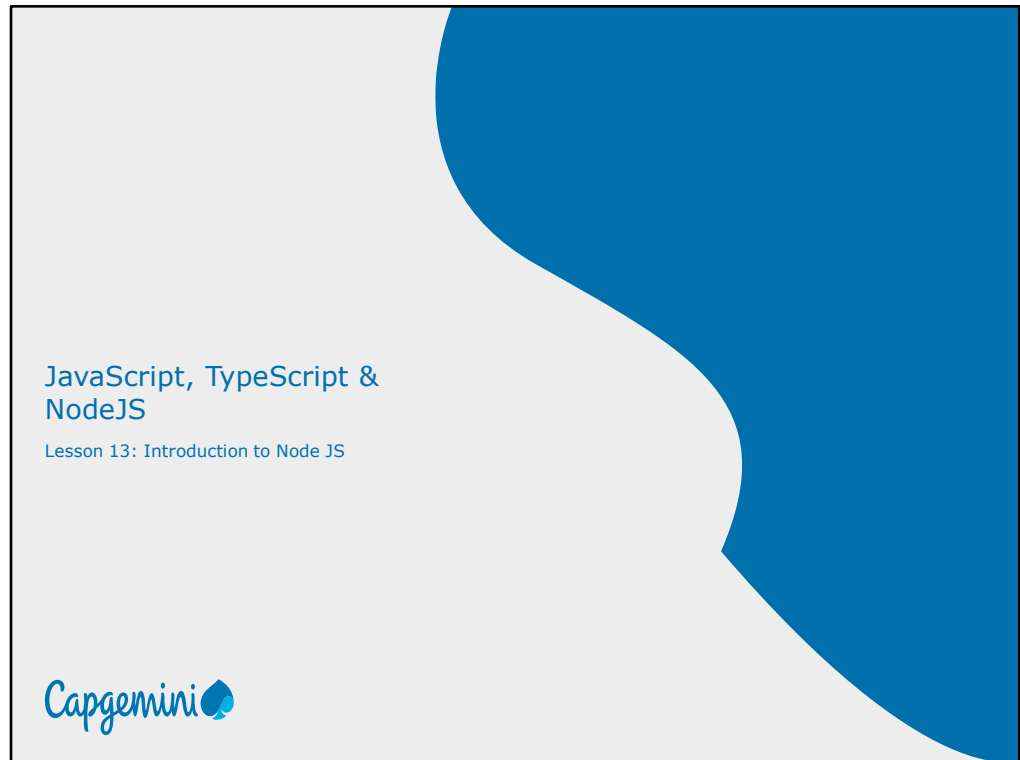




Lesson Objectives

- Introduction to Node js
- Installation of Node js
- Basics of Node js
- First Application of node js
- Node js NPM
- Node js module
- Node js Assertion module



13.1: JavaScript for Test Automation

Introduction to Node JS



- Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to execute JavaScript code outside of a web browser, typically on a server or a local machine.
- It is built on Google Chrome's V8 JavaScript engine, which compiles JavaScript into fast, efficient machine code.
- Key characteristics and features of Node.js include:
 - Server-side JavaScript:
 - Asynchronous and Event-Driven:
 - Single-threaded with Event Loop:
 - NPM (Node Package Manager):
 - Suitable for Real-time Applications:
 - Fast and Scalable:

13.2: JavaScript for Test Automation Installation of Node JS



- Download Node.js Installer for Windows
- Go to the site <https://nodejs.org/en/download/> and download the necessary binary files.

Download the Node.js source code or a pre-built installer for your platform, and start developing today.

LTS
Mature and Dependable

Stable
Latest Features


Windows Installer
node-v4.2.3-x86.msi


Macintosh Installer
node-v4.2.3.pkg


Source Code
node-v4.2.3.tar.gz

Windows Installer (.msi)	32-bit	64-bit
Windows Binary (.exe)	32-bit	64-bit
Mac OS X Installer (.pkg)		64-bit
Mac OS X Binaries (.tar.gz)		64-bit

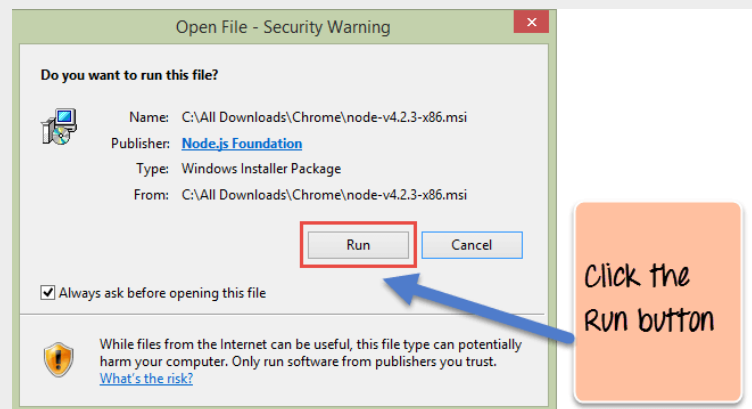
Download
the 32-bit
installer

13.2: JavaScript for Test Automation

Installation of Node JS



- Run the installation
- Double click on the downloaded .msi file to start the installation.
- Click the Run button on the first screen to begin the installation.

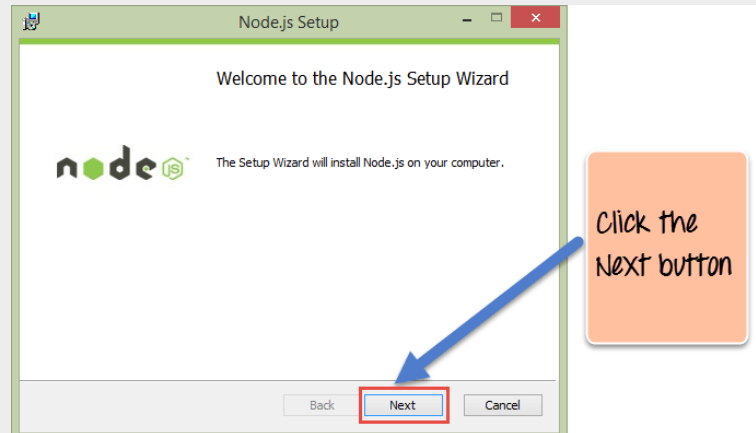


13.2: JavaScript for Test Automation

Installation of Node JS



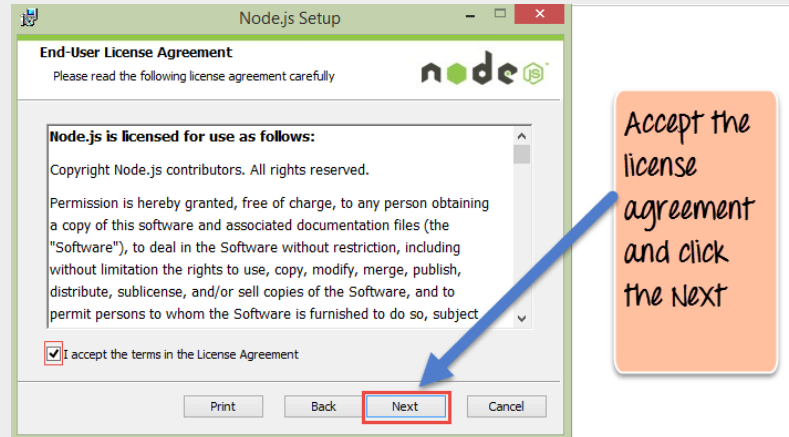
- Continue with the installation steps
- In the next screen, click the “Next” button to continue with the installation



13.2: JavaScript for Test Automation Installation of Node JS



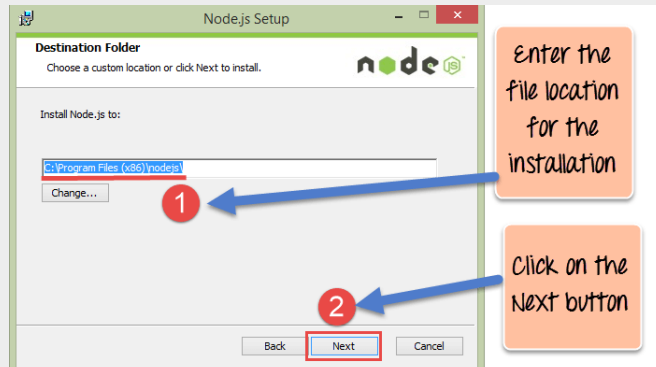
- Accept the terms and conditions
- In the next screen, Accept the license agreement and click on the Next button.



13.2: JavaScript for Test Automation Installation of Node JS



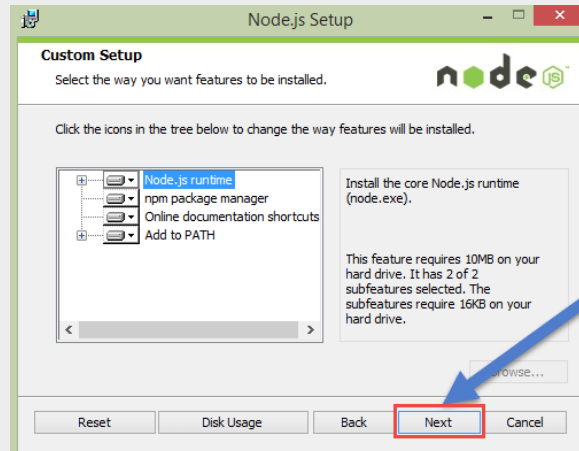
- Set up the path
- In the next screen, choose the location where Node.js needs to be installed and then click on the Next button.
- First, enter the file location for the installation of Node.js. This is where the files for Node.js will be stored after the installation.
- Click on the Next button to proceed ahead with the installation.



13.2: JavaScript for Test Automation

Installation of Node JS

- Select the default components to be installed
- Accept the default components and click on the Next button.



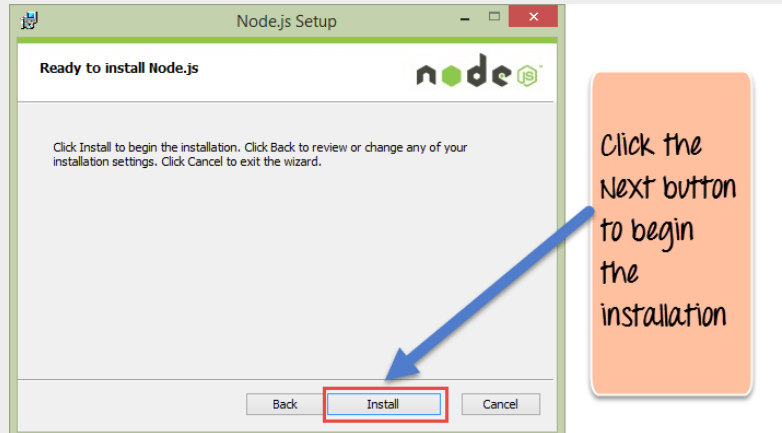
Accept the
default
components
and click on
Next

13.2: JavaScript for Test Automation

Installation of Node JS



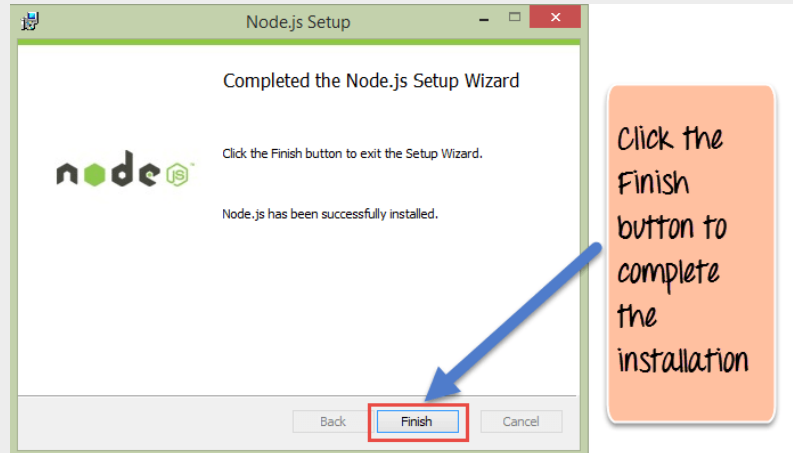
- Start the installation
- In the next screen, click the Install button to start installing Node.js on Windows.



13.2: JavaScript for Test Automation Installation of Node JS



- Complete the installation
- Click the Finish button to complete the installation.



13.3: JavaScript for Test Automation

Basics of Node JS



- Key Features
- **Single-threaded** → Handles one main thread but manages many connections efficiently.
- **Event-driven** → Executes tasks based on events (like requests, timers, or file operations).
- **Non-blocking I/O** → Does not wait for tasks like file reads or API calls to finish; continues executing other code.
- **Highly scalable** → Handles thousands of concurrent requests.

13.4: JavaScript for Test Automation

First Application of Node.js



- Once you have installed Node.js, let's create your first server that says "Hello Node.js!" in a web browser.
- Create a file called myfirst.js and add this code:

```
console.log("Hello Node.js!");
```

13.5: JavaScript for Test Automation

Node.js NPM



What is NPM?

- NPM is a package manager for Node.js packages, or modules if you like.
- www.npmjs.com hosts thousands of free packages to download and use.
- The NPM program is installed on your computer when you install Node.js

What is a Package?

- A package in Node.js contains all the files you need for a module.
- Modules are JavaScript libraries you can include in your project.

13.5: JavaScript for Test Automation

Node.js NPM



Download a Package

- Downloading a package is very easy.
- Open the command line interface and tell NPM to download the package you want.
- I want to download a package called "upper-case":

```
npm install upper-case
```

13.6: JavaScript for Test Automation

Node.js Module



- Node.js modules are self-contained units of code that encapsulate specific functionalities, promoting code reusability, organization, and maintainability in Node.js applications.
- They enable developers to break down large applications into smaller, manageable pieces, preventing global namespace pollution and improving overall code structure.
- In Short:
 - Modules = reusable files.
 - Export/Import example:

```
// math.js
exports.add = (a,b) => a+b;
// app.js
const math = require('./math');
console.log(math.add(2,3));
```


13.7: JavaScript for Test Automation

Node.js Assertion Module



- Node.js provides a built-in assert module that offers a set of assertion tests primarily used for verifying invariants and for debugging purposes within applications.
- While it can be used for basic testing, it is not a full-fledged testing framework.

```
const assert = require('assert');  
assert.equal(1+1, 2);
```

Review Question



Question 1: State True or False:
Node.js is built on which v7 engine?

Question 2: Which command checks the Node.js version installed?

- a) node version
- b) node -version
- c) node -v
- d) npm -v



Question 3: What is the role of NPM in Node.js?

- a) Runs Node.js applications
- b) Manages JavaScript packages and libraries
- c) Compiles TypeScript into JavaScript
- d) Connects Node.js to databases

Question 1 – False

Question 2 – Option C

Question 3 – Option B

Summary



In this lesson, you have learnt

- Introduction to Node js
- Installation of Node js
- Basics of Node js
- First Application of node js
- Node js NPM
- Node js module
- Node js Assertion module



Add the notes here.